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Milestones Supplement: Operationalizing Competency-Based Medical Education: How Programs Use the ACGME Milestones 2.0

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Abstract:	Background The Milestones 2.0 assessment framework was implemented by the Accreditation Council for Graduate Medical Education (ACGME) to support competency-based assessment of trainees. Objective This study sought to understand how the Milestones are being used by Graduate Medical Education (GME) programs nationally. Methods A validated survey instrument collecting respondent demographics, perceptions of the Milestones, and information regarding their use and implementation was distributed to Graduate Medical Education (GME) stakeholders nationally from January to March of 2025. Following a computer-assisted deductive analysis that used published, core principles of Competency-Based Medical Education (CBME) as initial codes, the authors completed an inductive thematic content analysis of responses. Results The survey received a total of 2,567 responses, including 1,306 complete responses. Due to the national nature of the survey, an exact response rate cannot be calculated. Most responses came from program directors (N=843) and non-physician administrative stakeholders (N=403). Based on the qualitative analysis, 13 themes were identified across the five core components of CBME: outcome competencies, tailored learning experiences, sequenced progression, competency-focused coaching, and programmatic assessment. While many themes emphasized use of the Milestones to support CBME principles, several practical and conceptual concerns emerged around the use of milestone data. Conclusions The themes that emerged through this analysis demonstrate that Milestones 2.0 provides an assessment framework for programs that aligns with the core CBME principles. At the same time, further work is needed to ensure that faculty and trainees can use the Milestones effectively and that determinations of competence correlate with practical clinical outcomes.

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Cover Letter

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Dr. Gail Sullivan, Editor-in-chief
Journal of Graduate Medical Education
Via electronic submission

Re: Operationalizing Competency-Based Medical Education: How Programs Use the ACGME Milestones 2.0

Dr. Sullivan,

Despite the requirement to report data using the Accreditation Council for Graduate Medical Education (ACGME) Milestones 2.0 framework, little is understood about how programs use this data internally to promote Competency-Based Medical Education (CBME). The attached manuscript, *Operationalizing Competency-Based Medical Education: How Programs Use the ACGME Milestones 2.0*, leverages qualitative data from a recent national survey to assess how programs use this data for CBME.

This manuscript is submitted as a component of the upcoming supplement around the ACGME Milestone 2.0 framework. We believe the Graduate Medical Education community, your primary readership, will be personally and professionally interested in how their colleagues use these data.

We confirm that all authors have met all four criteria for authorship in the ICMJE Recommendations. No medical writers or artificial intelligence systems were involved in the writing process. All authors accept responsibility for this publication. We confirm that this manuscript is not being considered by any other journal and has not been published elsewhere, including in abstract form.

We thank the editorial staff and potential reviewers in advance for any feedback. We look forward to discussing this manuscript with you further in the coming days.

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Milestones Supplement: Operationalizing Competency-Based Medical Education: How Programs Use the ACGME Milestones 2.0

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Abstract

Background: The Milestones 2.0 assessment framework was implemented by the Accreditation Council for Graduate Medical Education (ACGME) to support competency-based assessment of trainees.

Objective: This study sought to understand how the Milestones are being used by Graduate Medical Education (GME) programs nationally.

Methods: A validated survey instrument collecting respondent demographics, perceptions of the Milestones, and information regarding their use and implementation was distributed to Graduate Medical Education (GME) stakeholders nationally from January to March of 2025. Following a computer-assisted deductive analysis that used published, core principles of Competency-Based Medical Education (CBME) as initial codes, the authors completed an inductive thematic content analysis of responses.

Results: The survey received a total of 2,567 responses, including 1,306 complete responses. Due to the national nature of the survey, an exact response rate cannot be calculated. Most responses came from program directors (N=843) and non-physician administrative stakeholders (N=403). Based on the qualitative analysis, 13 themes were identified across the five core components of CBME: outcome competencies, tailored learning experiences, sequenced progression, competency-focused coaching, and programmatic assessment. While many themes emphasized use of the Milestones to support CBME principles, several practical and conceptual concerns emerged around the use of milestone data.

Conclusions: The themes that emerged through this analysis demonstrate that Milestones 2.0 provides an assessment framework for programs that aligns with the core CBME principles. At the

same time, further work is needed to ensure that faculty and trainees can use the Milestones effectively and that determinations of competence correlate with practical clinical outcomes.

Introduction

Educational Milestones were implemented for the assessment of graduate medical trainees by the Accreditation Council for Graduate Medical Education (ACGME) in 2013.[1,2] One goal of these milestones was to continue the shift in trainee assessment toward a competency-based medical education (CBME) framework.[3,4] The Milestones assess trainees along an expected developmental trajectory across the six core competencies: interpersonal and communication skills, medical knowledge, patient care, practice-based learning and improvement, professionalism, and systems-based practice.

After the initial Milestones were developed, the ACGME underwent an iterative review process, creating a revised set of milestones termed "Milestones 2.0."[2] Milestone data are reported to the ACGME nationally for accreditation, but little is known about how programs use these data to operationalize CBME in practice. The core components of CBME[5] provide a framework for examining how Milestones inform: outcome competencies, sequenced progression, tailored learning experiences, competency-focused coaching, and programmatic assessment. This study analyzed the results from a national survey of stakeholders in Graduate Medical Education (GME) to understand how the Milestones 2.0 framework is being used to promote the core components of CBME.

Methods

Survey Design and Distribution

The survey on which this study is based was developed by Laura Hirshfield of the University of Illinois Chicago and the ACGME Milestones development team. The initial phase of their development was based on interviews with graduate medical education leaders (program directors, Clinical Competency Committee [CCC] chairs, etc.) across ACGME-accredited specialties. These interviews were conducted by trained interviewers, recorded, and transcribed; an inductive analysis of the transcripts was then used to develop the final survey. The full methods used in the development of this survey are concurrently reviewed and published alongside this article (L. E. Hirshfield, P. B. Barlow, I. Haynes, S. McLean, L. Edgar, unpublished manuscript, 2026) . The final iteration of the survey contained 34 questions, including multiple-choice, Likert-scale, and open-ended, narrative items. The survey was distributed via Qualtrics XM (Qualtrics, Provo, UT) to ACGME-affiliated stakeholders nationally from January 30 – March 30, 2025.

Data Analysis

Descriptive data regarding response rates and respondent demographics were summarized using STATA Version 19.5 (StataCorp LLC, College Station, TX). The authors reviewed the survey and coded each question according to the CBME component(s) it addressed to ensure there was comprehensive representation of all core components. The core components framework is from published work by Van Melle et al and includes: outcome competencies, sequenced progression, tailored learning experiences, competency-focused coaching, and programmatic assessment.[5] Given the volume of qualitative data, Computer-Assisted Qualitative Data Analysis (CAQDAS) using MAXQDA 24 (VERBI Software, Berlin, Germany) was used for initial data coding. Based

on author consensus after reviewing the survey, an initial deductive analysis coded the data using established core components of CBME from the literature as the initial codes.[5]

The coded responses from the initial deductive analysis were then reviewed by the authors in pairs: outcome competencies (SM, RM), tailored learning experiences and sequenced progression (IH, SA), and finally competency-focused coaching and programmatic assessment (SAM, MB). Each author pairing independently reviewed the responses coded to their core components through thematic content analysis, identified representative quotes (reported throughout by randomly assigned respondent numbers in parentheses), and then met to resolve discrepancies until each pair reached thematic saturation. The author group then met to review all themes. When similar themes were identified across multiple core components, the authors debated until consensus was reached on which core component most appropriately related to the theme. This process was repeated until the authors felt no further key themes or concepts could be drawn from the dataset.

The Institutional Review Board of the University of Illinois Chicago reviewed the survey process and determined that it did not constitute human subjects research as defined under United States law in 45 C.F.R. § 46.

Results

There were a total of 2,567 responses to the survey including incomplete or partially answered surveys (N=1306 complete surveys). The survey was distributed to individuals affiliated with the over 14,000 ACGME-accredited training programs and distribution to other stakeholders was encouraged. Due to this referral-based, snowball sample of GME stakeholders, a conventional

response rate cannot be calculated. The roles most commonly disclosed by respondents were that they were program directors (N=843/1600, 52.7%) or in non-physician administrative or leadership roles (N=403/1600, 25.2%). The largest proportion of respondents had been in their current role for less than five years (N=812/1675, 48.5%). Respondent demographics are summarized in Table 1.

		N (%)
What is your primary role in GME?		
Program Director		843 (52.7)
Associate/Assistant Program Director		165 (10.3)
Physician Faculty		93 (5.8)
Non-physician Faculty		5 (0.3)
Designated Institutional Official (DIO)		41 (2.6)
Resident/Fellow		50 (3.1)
Non-physician Administration/Leadership		403 (25.2)
Which of the following best describes your program? *		
Academic		1261 (65.5)
Community		460 (29.3)
Rural		85 (4.4)
Federally Qualified Health Center (FQHC)		68 (3.5)
Veteran Affairs (VA)		51 (2.6)
If applicable, approximately how many of the following are in your program?		N (%)
Residents	0	167 (13.4)
	<10	119 (9.5)
	10-19	330 (26.5)
	20-29	274 (22.0)
	30-39	143 (11.5)
	40-49	87 (7.0)
	50-59	28 (2.2)
	60+	99 (7.9)
Fellows	0	453 (35.1)
	1-2	261 (20.2)
	3-4	172 (13.3)
	5-7	151 (11.7)
	8-10	109 (8.4)
	>10	144 (11.2)
How many years have you been in your current role?		
<5 years		812 (48.5)
5-10 years		520 (31.0)
>10 years		343 (20.5)

Table 1. Descriptive statistics of respondents' self-reported demographics. Note that multiple program types (*) could be selected by respondents. Not all respondents answered all demographic questions. GME = Graduate Medical Education.

Respondents were explicitly asked whether different stakeholders use their milestone data. Of respondents, most stated that learners (N = 179/283, 63.3%) and core faculty (N = 178/283,

62.9%) use their Milestones 2.0 data. The majority of frontline faculty, however, were thought to not use Milestones 2.0 data (N = 173/281, 61.6%).

	Yes	No	Unsure
	N (%)	N (%)	N (%)
Do _____ use their Milestones data?			
Learners	179 (63.3)	52 (18.4)	52 (18.4)
Core Faculty Members	178 (62.9)	72 (25.4)	33 (11.7)
Frontline Faculty Members	49 (17.4)	173 (61.6)	59 (21.0)

Table 2. Descriptive quantitative statistics regarding use of the Milestones 2.0 data. Note that not all respondents answered all questions.

Across core components, respondents described the Milestones as providing a framework that can be used for assessment, coaching, individualized learning, and curriculum review while also identifying challenges in translating that framework into consistent, meaningful practice. These concerns reflected dissatisfaction with the design of the Milestones, rather than rejection of CBME as an educational philosophy. One respondent captured this sentiment, stating that “I do think we need Milestones and competencies to guide our evaluations and believe that CBME is the correct approach, I just don't think [the current framework] really gets at that intention in a clear, concise, user-friendly way” (Respondent 66). The results of the below thematic analysis show how this tension shaped Milestone use across the five core components of CBME.

Outcome Competencies

Shared Outcome Expectations

Respondents described the Milestones as providing a shared framework for defining the competencies expected of residents and fellows throughout their training. Standardized language and developmental expectations helped faculty and learners establish a common understanding of physician competence and what successful performance should look like across stages of training. One respondent described the Milestones as "a guide that we can share with fellows and faculty to have common language and expectations" (12) by "help[ing] the faculty create a shared mental model" (183). Respondents also valued the breadth of the framework, describing it as encompassing competencies extending beyond medical knowledge and patient care.

However, respondents questioned whether these outcomes were consistently defined with sufficient specificity to meaningfully represent competence. Although Milestones "describe a broad array of competencies" (10), another described them as "too nebulous and not tied to actual observable behaviors" (24). Respondents valued a common framework for defining outcomes while expressing uncertainty about how individual Milestones match observable clinical outcomes.

Alignment with Independent Practice

Responses emphasized using the Milestones to evaluate learners' progression toward autonomous, unsupervised practice and to support graduation decisions. Some viewed the breadth of the Milestones as supporting this goal, noting that their "broad scopes allow programs [to] assess the multitude of skills, qualities, and abilities needed to practice independently" (23). However,

respondents questioned whether all Milestones "…translate well to the requirements of independent practice" (54). These responses highlighted a tension between the value of a comprehensive framework for defining competency outcomes and concern about whether some Milestones appropriately reflect the capabilities required of practicing physicians.

Sequenced Progression

Defining the Typical Developmental Trajectory

Respondents described the Milestones as defining an expected developmental trajectory for trainees. One respondent described the Milestones as providing "meaningful levels that clearly differentiate between average, good and excellent trainees" (120). Others noted that they allow programs to clearly "understand where the learner is in the development to independence" (125). These responses emphasized that the framework provides a shared reference for understanding learner progression, setting developmental goals, and identifying trainees who begin to "fall off the [expected developmental] curve" (43).

Challenges Distinguishing Between Performance Levels

Although many respondents viewed the Milestones as clearly defining developmental progression, others felt that there is "indistinct differentiation within and between levels" (98). Responses described difficulty translating Milestone descriptors into observable behaviors, resulting in inconsistent interpretation and scoring. These challenges were reported to be particularly pronounced in non-patient-facing specialties and specialties that perceived limited alignment with the harmonized Milestones. Across both of these groups of respondents, there were reports that it was challenging for "faculty to use the milestones as a progression along a training pathway" (15).

Tailored Learning Experiences

Learner-Centered Experiences

Programs used Milestone data to individualize learning experiences based on trainees' strengths, areas for growth, and developmental goals. Respondents described using Milestones to develop individualized learning plans (ILPs), tailor remediation strategies for struggling learners, and identify trainees ready for greater autonomy, leadership opportunities, or advanced educational experiences. They allow CCCs to "focus on the outliers as well to provide more meaningful individualized plans," (41) ensuring high-performing trainees are "offered additional experiences and more independence" (138). To complete the sufficiently detailed assessments needed to design these tailored experiences, programs have felt increased “administrative, clinical, and documentation demands [that] contribute to burn out” (81). While programs are working to create learning experiences that are learner-centric, this administrative burden can stifle these efforts.

Competency-Focused Coaching

Individualized Learner Development

Respondents described incorporating Milestones into advisory meetings, coaching sessions, and semiannual reviews to promote learner reflection, compare self-assessments with faculty evaluations, and identify strengths and areas for growth. Some programs reported assigning trainees a "faculty advisor” (85) or “clinical coach who reviews [learners’] assessments with them” (87). Respondents used discrepancies to guide coaching conversations and establish individualized goals, for example, by “understand[ing] how (and why) discrepancies might arise between faculty (CCC) evaluations and self-evaluations” (140).

Coaching Across the Performance Spectrum

Programs use milestone data to tailor coaching for learners across the performance spectrum, although respondents described substantially greater emphasis on remediation than on development for high-performing trainees. One respondent explained that the Milestones allowed faculty to "further drill down" (30) and identify specific developmental needs. In contrast, several respondents found that the Milestones "generally don't provide useful feedback for high-performing learners" (20) and were "less helpful for residents that are overall doing well" (44).

Integration with Local Coaching Frameworks

Respondents described adapting the Milestones to local educational models that better aligned with specialty-specific curricula and coaching processes. Examples included “creat[ing their] own anchors for the milestones” (61) or mapping Milestones back onto more familiar EPAs to help faculty with the transition to this assessment framework.

Barriers to Competency-Focused Coaching

Respondents identified three primary barriers to using Milestones for competency-focused coaching: insufficient specificity, inconsistent faculty interpretation, and limited learner engagement. Many described the Milestones as too broad to support actionable coaching because they "lack sufficient granularity for meaningful feedback" (20). Variability in faculty interpretation reduced the consistency of Milestone ratings, with one respondent describing them as "too vague and subjective amongst individuals and programs" (29). Finally, respondents stated that learners value narrative feedback and coaching conversations more than Milestone reports, with some

reporting that residents "don't take the milestones as an important measure of their educational progress" (3).

Programmatic Assessment

Assessment of Trainee Competence

Programs use milestone data to assess trainee competence longitudinally and support decisions regarding learner progression and graduation. Respondents described using Milestones to monitor individual developmental trajectories, compare performance within and across cohorts, and determine readiness for unsupervised practice. Milestones were viewed as "an objective framework with which to assess learners" (92) in order "to ensure that graduates are at an [appropriate level] prior to graduation" (1). Respondents also emphasized that Milestone ratings were intended to document attainment of competency rather than completion of educational experiences through "consistent information that supports a level of competence" (59).

Integration Within Broader Assessment Systems

Respondents described integrating Milestone data with workplace-based assessments, faculty evaluations, peer assessments, self-assessments, EPAs, portfolios, and other assessment data to inform CCC decisions. This allows programs to complete "a global review of each trainee in assessing the trainee's progress" (126). Some respondents felt, however, that integration of these assessments is too “laborious” (13) and imposes too heavy an administrative load.

Curricular Development

Programs also used aggregate Milestone data to evaluate curricula and identify opportunities for program improvement. Respondents described reviewing cohort-level trends to “identify... gaps in the curriculum” (16), determine where additional educational experiences were needed, and evaluate the effectiveness of their training programs. In doing so, programs can identify “curricular holes” (32) and determine competencies that trainees "may need more exposure [to] earlier or later" (52).

Challenges with Programmatic Assessment

Respondents identified several barriers to using the Milestones for programmatic assessment, including concerns regarding validity, reliability, specialty-specific applicability, and administrative burden. Some questioned whether Milestone ratings reflected meaningful educational or clinical outcomes, reporting that "the milestones themselves don't translate to better patient outcomes" (30). Others see substantial variability in faculty interpretation, noting that some faculty used Milestones "as a Likert scale of bad to good, rather than progression along a training pathway" (32). Respondents also described difficulty assessing competencies such as systems-based practice (SBP) and practice-based learning and improvement (PBLI), questioned the applicability of Harmonized Milestones across specialties, and reported that administrative burden reduced faculty engagement and the perceived value of the assessment process. Respondents referred to the Milestones as "an added admin[istrative] burden on an already overwhelming job” (50). One respondent summarized this challenge by stating that "it is very hard to gain and maintain buy in and participation in this process" (88).

The themes derived from the inductive qualitative analysis are summarized in Table 3 below with representative quotes for each theme.

Core Component	Themes	Representative Quotes (Respondent ID)
Outcome Competencies	Shared Outcome Expectations	Milestones are "a clear way to understand what is expected of residents as they progress through their program and they help the faculty create a shared mental model for evaluating residents." (Respondent 183)
	Readiness for Independent Practice	Milestones are useful for "preparing trainees to function on an independent level," (79) "advancing residents toward autonomy and competence," (36) and "accurately document[ing] a resident is ready for independent practice." (134)
Sequenced Progression	Defining the Typical Developmental Trajectory	The "granularity and stepwise graduated fashion of Milestones 2.0" (116) "allow the CCC and PD to really understand where the learner is in the development to independence." (170)
	Challenges Distinguishing between Performance Levels	The developmental levels are "vague and not entirely distinguishable." (63)
Tailored Learning Experiences	Learner-Centered Experiences	Milestones allow the CCC to “provide more meaningful individualized plans.” (116)
Competency-Focused Coaching	Individualized Learner Development	A "faculty advisor reviews milestones with [the] resident and develops a learning plan." (182)
	Coaching Across the Performance Spectrum	Milestones allow faculty to "further drill down" (50) to identify specific developmental needs in struggling trainees but are "less helpful for residents that are overall doing well." (123)
	Integration with Local Coaching Frameworks	"The milestones are mapped back onto the EPAs which make more sense to our faculty and residents." (50)
	Barriers to Competency-Focused Coaching	Milestones are "too broad and not useful for residents to know how to improve” (110) and “subjective amongst individuals and programs." (56) “Most residents don't care to even review their milestones making them ineffective." (126)
Programmatic Assessment	Assessment of Competence	Milestones are "an objective framework with which to assess learners" (119) that provides "consistent information that supports a level of competence." (87)
	Integration Within Broader Assessment Systems	"Milestones used in conjunction with other assessment tools i.e., operative evaluations, 360s, etc., seem to help our faculty make informed decision about progressive autonomy." (89)
	Curricular Development	Milestones "regularly illustrate curricular holes and that helps us move forward.” (87)
	Challenges with Programmatic Assessment	There is "no correlation between the [Milestones] and how good of a doctor someone is." (60) Ratings are often assigned "based on year instead of what the competencies actually direct.” (23) Milestones are "more busy work than anything," (71) "way too cumbersome," (81) “essentially a waste of everyone’s time,” (17) or "an added admin burden on an already overwhelming job." (80) "It is very hard to gain and maintain buy in and participation in this process.” (155)

Table 3. Themes identified through inductive, thematic analysis of survey responses identified by MAXQDA as relevant to the core components of Competency-Based Medical Education.[5] (Abbreviations: CCC = Clinical Competency Committee; PD = Program Director)

Discussion

CBME is an outcomes-based approach to medical training, structured around an organizing framework of competencies. For GME in the United States, the organizing framework for training programs is the ACGME Milestones 2.0.[2,9] The expectation is that every trainee progresses through the Milestones and ultimately achieves competence in the unsupervised care of patients and families within their specialty by the completion of their training program.[6–8] Our findings demonstrate that programs use the Milestones in ways that align with all five core components of CBME,[5] providing faculty and residents with a shared mental model of expectations and a structure for translating competency-based principles into educational practice,[10] albeit with several identified limitations.

The themes representing successful use of the Milestones described here align with existing literature describing CBME through lenses of: learner-centered coaching and growth,[11–13] progression along a performance continuum,[14–17] and the use of educational data to evaluate programs and learning experiences.[18–22] Our findings further demonstrate how Milestones are used across these interconnected functions: to define shared expectations for competence, establish developmental trajectories, assess readiness for unsupervised practice, individualize learning and coaching, support progression and graduation decisions, and identify opportunities for curricular and program improvement.

Throughout the analytical process, the authors noted that the identified themes often reflected multiple core components of CBME. The theme of "readiness for independent practice," as an example, was often described in the context of the competency outcome but also informed

longitudinal assessment of progression, identification of learners requiring additional support, and tailored coaching or experiences that move trainees toward competency. Similarly, aggregate Milestone data were used not only to assess individual learners but also to identify curricular gaps and evaluate training programs. These findings illustrate that the core components of CBME function as interconnected pieces rather than discrete processes. The Milestones may serve not only as a required measure of outcomes, but also as an organizing framework that links these components in educational practice.

While there are productive efforts to use the Milestones to support CBME, there were also multiple reported limitations of the Milestones. Across themes, these limitations relate to uncertainty regarding how faculty or trainees should use milestone data, which may contribute to decreased confidence in Milestones 2.0 as an assessment tool. Several of these limitations appeared across more than one component, suggesting that limitations in the assessment framework may constrain the usefulness of Milestones for CBME in some circumstances. Multiple respondents did, however, distinguish support for CBME as an educational philosophy from their dissatisfaction with their current use of the Milestones.

While the authors believe this analysis supports the growing literature on the benefit of the Milestones as an assessment strategy,[14,15,23] we recognize that there remain questions regarding the long-term implications of Milestone-based assessment.[24] Further faculty and programmatic development at national and institutional levels is needed to guide the use and interpretation of Milestones in educational practice. These efforts should, in part, focus on the Milestones as an

initially formative and ultimately summative assessment of clinical competence and readiness to practice safely without supervision.

Additionally, ongoing research efforts will be necessary to continually validate the Milestones as markers of clinical competence. Such work should examine whether Milestone ratings are associated with meaningful educational and clinical outcomes and whether their use improves learner development or performance to address some of the concerns raised by respondents in this survey.[25] This work should be careful not to conflate clinical competence with the notion of clinical excellence, which merits its own line of research and is a nuanced discussion beyond the scope of this paper.[26]

This work, in addition to the aforementioned, is primarily limited by the potential for interpretive bias on the part of the authors and the over-generalized nature of these findings across specialties and programs. To limit interpretive bias, authors were intentionally divided into groups to limit institutional overlap, and all themes were discussed both in author pairs and with the entire author group to ensure thematic saturation and consistency in interpretation. As pointed out by some respondents, the Milestones may also be applied differently in different specialties even after attempts to make them widely applicable. However, this analysis lacked adequate data to fully characterize these differences. The qualitative nature of the study also describes how stakeholders understand and report using Milestones but cannot establish the frequency of these practices or their effects on learner or program outcomes. Further work should focus on quantitative, learner- and specialty-centered educational outcomes to aid in interpreting these qualitative findings and address these limitations.

Conclusions

This study provides further support that GME stakeholders nationally are utilizing the Milestones 2.0 assessment framework in alignment with the core components of CBME to establish shared expectations for competence, characterize developmental progression, individualize learning and coaching, inform progression and graduation decisions, and guide programmatic improvement. These overlapping uses suggest that the Milestones can function as more than an outcomes-based assessment tool; they can provide an organizing framework through which programs operationalize interconnected components of CBME. However, limitations in interpretation, variability in use, and administrative burden may constrain their effectiveness. Further research is needed to quantify these findings, and further educational development is needed to improve stakeholder uptake of evidence-based principles of Milestone utilization.

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Supplemental Material_Survey
Click here to access/download;Supplemental Material;FINAL Milestones Survey Jan 2025.docx

Milestones 3.0 Survey

Introduction

Thank you for taking the time to provide your experience with Milestones 2.0. The Milestones 2.0 development process concluded in 2023. Efforts by the ACGME then moved to conducting a principles-focused program evaluation in collaboration with Dr. Laura E. Hirshfield, PhD, at University of Illinois Chicago (UIC). Dr. Hirshfield has developed a survey guided by the principles upon which the Milestones were first developed and interviews conducted with 45 stakeholders. The survey is designed to elicit your perspectives and experiences regarding the implementation and use of Milestones 2.0. Only aggregated summary results will be shared with the ACGME.

As a user of the Milestones, we invite you to participate in helping the ACGME identify and understand how the Milestones are, or are not, working within your context. By keeping the survey anonymous, we strongly encourage you to provide your honest feedback to improve the Milestones framework to better serve the GME community and its learners.

You may not be familiar with one phrase used within the survey – harmonized Milestones. Harmonized Milestones include a set of 11 subcompetencies across Interpersonal and Communication Skills, Practice-Based Learning and Improvement, Professionalism, and Systems-Based Practice that describe knowledge, skills, and behaviors applicable across all surgical and medical specialties. Each specialty was offered the opportunity to make specialty-specific edits. Few specialties made substantial changes to the harmonized Milestones.

The UIC Institutional Review Board determined that this evaluation does not constitute research involving human subjects as defined by federal guidelines.

The survey should take approximately 15 minutes to complete.

Section 1: Demographics

1. What is your primary role in GME? (check one)
    a. Program director
    b. Associate/assistant program director
    c. Physician faculty
    d. Non-physician faculty
    e. Designated institutional official (DIO) *(If selected, skip Q3, 5,7, 8)*
    f. Resident/fellow
    g. Non-physician administration/leadership (e.g., coordinator, nurse)
    h. Other (please specify):_____
2. How many years have you been in your current role?
3. For the purposes of this survey, please focus on one specific specialty/subspecialty for your responses. What residency or fellowship program are you responding for? (check one/drop down)
4. Which of the following professional degrees have you earned? (Check all that apply)
5. If applicable, approximately how many of each the following are in your program?
    a. Residents:

b. Fellows:
c. Preliminary residents:
Please check here if the above are estimates ☐

6. Which of the following best describes your program? (check all that apply)
    a. Academic (i.e. hospitals that are affiliated with medical schools and focus on education and research)
    b. Community (i.e., non-academic hospitals that provide a range of services to local communities)
    c. Rural
    d. Federally Qualified Health Center (FQHC)
    e. Veteran Affairs (VA)
    f. Uncertain
    g. Other (please specify): _____
7. What is/was your role(s) with your program's Clinical Competency Committee (CCC)? (select any that apply, past or present)
    a. Member
    b. Chair
    c. Non-voting member
    d. Chief resident member
    e. Not a member
8. Please specify (EXCLUDE IF SELECT 7e)
    a) What other specialty(ies) or subspecialty(ies) does your CCC review?
    b) How many members serve on your program's CCC (including yourself)?
    c) How many non-physician members serve on the CCC?
    d) How many times per year does your CCC meet?
    e) Does your CCC receive ongoing faculty development? y/n

Section 2: Opinions

For the following questions, please only answer based on the residency or fellowship you selected in the previous section. For DIOs, please answer based on your overall experience across specialties.

1. How important is it that all learners of the same specialty are assessed against the same benchmarks outlined in the Milestones?

    Not at all important | Somewhat important | Important | Very important | Extremely important

    If you selected "not at all important" or "somewhat important," please elaborate:

2. How satisfied are you with Milestones 2.0 as a tool for supporting the adoption of competency-based medical education (CBME) components (assessment, curriculum, feedback, programmatic assessment, individualized learning plans, etc.) for each of the following?
    a. All six Core Competencies
    b. Patient Care (PC)
    c. Medical Knowledge (MK)
    d. Systems-Based Practice (SBP)
    e. Practice-Based Learning and Improvement (PBLI)
    f. Professionalism (PROF)
    g. Interpersonal and Communication Skills (ICS)

    Very unsatisfied | Unsatisfied | Satisfied | Very satisfied | Unsure

    If you selected "very unsatisfied" or "unsatisfied," please elaborate:____

3. How effectively do Milestones 2.0 support each of the following?
    a. Program improvement goals (e.g., increased participation in quality improvement activities)
    b. Educational outcomes for learners (e.g., higher in-training exam scores)
    c. Learner self-assessment (i.e., resident/fellow may perform informed self-assessment)
    d. Patient outcomes
    e. Individualized learning plans (e.g., customized set of activities to achieve personalized educational goals)

    Not at all effectively | Somewhat effectively | Effectively | Very effectively | Unsure

    If you selected "not at all effectively" or "somewhat effectively," please elaborate:____

4. How satisfied are you with the following aspects of Milestones 2.0?
    a. Simple language and design.
    b. Distinction between meaningful levels of competency (e.g., between Milestone level 2 and 3)
    c. Representation of realistic progression of knowledge, skills, and behaviors (i.e., across the 5 levels)
    d. Supplemental Guide (ACGME companion document to Milestones 2.0)
    e. Harmonized Milestones that were developed for all specialties
    f. Representative of national standards

    Very unsatisfied | Unsatisfied | Satisfied | Very satisfied | Unsure

Please elaborate, as desired:

5. How effectively do harmonized Milestones foster consistent curriculum and assessment across each of the following?

    a. Institutions

    b. Departments
    c. Specialty societies/boards

Not at all effectively | Somewhat effectively | Effectively | Very effectively | Unsure

Please elaborate, as desired:

6. How effectively do the content and competencies assessed by Milestones 2.0 measure the knowledge, skills, and behaviors required by learners to serve the public without supervision?
   Not at all effectively | Somewhat effectively | Effectively | Very effectively | Unsure

    Please elaborate, as desired:

7. How effectively did Milestones 2.0 promote program improvement of the following?

    a. Assessment of learners

    a. Feedback to learners
    b. Individualized learning plans
    c. Curriculum for harmonized Milestones (PROF, PBLI, ICS, and SBP)

Not at all effectively | Somewhat effectively | Effectively | Very effectively | Unsure

Please elaborate, as desired:

8. How satisfied are you with the following aspects of Milestones reporting:

    a. Frequency (i.e., reporting Milestones twice per year)

    b. Timing (i.e., reporting Milestones October-January and April-June)

Very unsatisfied | Unsatisfied | Satisfied | Very Satisfied | Unsure

Please elaborate, as desired:

Section 3: Open-Ended Feedback

9. What aspects of Milestones 2.0 are most effective in supporting assessment and learner development? (Open text response)

10. What challenges have you encountered in implementing/ongoing use of Milestones 2.0? (Open text response)

11. What additional tools or resources would help you better use Milestones 2.0? (Open text response)

12. What, if any, faculty development is offered at your program regarding Milestones and/or assessment?

13. Do you have any additional feedback or suggestions for improving the Milestones? (Open text response)

14. Are you willing to take an additional 10-15 minutes to answer a few more in-depth questions regarding the milestones? Y/N (*yes go to section 4, no end survey*)

    If Y, select the type of questions you want to answer (check all that apply)

    a. Data use (two questions)
    b. Implementation and assessment (five questions)
    c. Clinical Competency Committee (CCC) (five questions)

Section 4: Additional questions for those who selected 14Y

Data Use

1. Do ___ use their Milestones data? Y/N/unsure matrix
   a. Learners
   b. Faculty members (core)
   c. Faculty members (frontline)
   If yes, please elaborate as desired:

2. Does your Program Evaluation Committee use Milestones data? If so, how? (Y/N/Unsure) Explain:

Implementation and Assessment

3. How are Milestones 2.0 used to identify and support the learner who is struggling?
4. How are Milestones 2.0 used to identify and support the accelerated learner?
5. On average, how many assessments per resident per competency are performed every six months?
6. Do the Milestones support equitable evaluation of all learners? If so, how or how not? (Y/N) Explain:
7. What does effective implementation of Milestones look like?

Clinical Competency Committee (CCC) Questions:

1. Describe the processes used by the CCC to assign milestones semi-annually, including data sources.
2. What is the role of your program director on the CCC?
   a. CCC chair

    b. CCC member
    c. CCC observer
    d. Program director does not attend
3. Do you have a patient/public representative on your CCC? y/n
    a. If yes, describe their role and how they participate_
4. What percentage of residents/fellows does your CCC review at each meeting?
    a. Less than 25 percent
    b. 26-50 percent
    c. 51-75 percent
    d. 76-100 percent
    e. Unsure
5. Who provides feedback from the CCC to the resident/fellow?
    a. Program director
    b. Assistant/associate program director
    c. CCC member (not program director or assistant/associate program director)
    d. Resident/fellow advisor